USN | | | | | | | | | | [23MSCH3H01]

THIRD SEMESTER UNIVERSITY EXAMINATION, JANUARY - 2025

SPECTROSCOPY - II

Time: 03 HOURS **Maximum Marks:50**

		Blooms Level	CO	Marks
	SECTION – A			
1.	**Answer any FOUR of the following questions.**			**4×5=20**
a.	List the differences between ESR and NMR spectroscopic techniques.	1	1	**05**
b.	The absorption and emission spectra of anthracene follow the mirror image rule. Explain.	2	2	**05**
c.	What is the effect of magnetic field on isomer shift of ^{57}Fe?	3	3	**05**
d.	Explain "improper axis of rotation" symmetry element and symmetry operation with an example.	2	5	**05**
e.	Explain the mathematical representation of "the great orthogonality theorem"	2	6	**05**
	SECTION – B			
2.	**Answer any TWO of the following questions.**			**2×9=18**
a.	What is the difference between the Non-rigid Rotor model and Rigid Rotor model? Derive the expression for rotational energy for a linear diatomic molecule using Non-rigid Rotor model. Draw the rotational energy levels and frequency lines.	3	3	**09**
b.	Explain the theory of Raman scattering. Draw the schematics of Raman scattering processes.	4	4	**09**
c.	Describe the matrix representation for clockwise rotation operation through an angle θ for Z-axis.	2	5	**09**
	SECTION – C			
3.	**Answer the following question.**			**1x12=12**
a.	**i.** Calculate the ESR frequency in a magnetic field of 25000 Gauss, for $g = 2$ and $\beta = 9.273 \times 10^{-24}\ JT^{-1}$.	2	1	**04**
	ii. Express the equation of overlap integral in FRET and discuss its significance.	2	1	**04**
	iii. Explain the quadratic functions x^2, y^2, z^2, xy, xz and yz in the character table of C_{2v} point group.	2	6	**04**